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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Scuba Adventurers
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG8, 9, 10,12, 12.3 12.5
Project / Solution Name	ExpiryGuardAI- Food Waste Prevention Assistant
Name (Leader)	Yap Sze Min
Email & Phone (Leader)	szemin3128@1utar.my, 013-5293128
University Affiliation(s)	Universiti Tunku Abdul Rahman
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Food waste is a persistent challenge for small and medium-sized enterprises (SMEs) in the food industry, including cafés, restaurants, food vendors, and cloud kitchens. These businesses rely on fresh ingredients to prepare meals, but managing inventory and tracking expiry dates can be difficult, especially when resources, staff, and time are limited.

Without effective inventory monitoring and timely action, ingredients may expire before being used, resulting in unnecessary food disposal and financial losses. Overstocking, unpredictable customer demand, poor stock rotation, and limited visibility into ingredient expiry dates can further worsen the problem. For small businesses operating on tight budgets, these losses can reduce profit margins and make daily operations less efficient.

The consequences extend beyond individual businesses. Discarded food wastes the water, energy, labour, and other resources used in its production and transportation. When food waste decomposes in landfills, it can also contribute to greenhouse gas emissions, intensifying environmental concerns.

Many SMEs may lack access to affordable, practical tools that help them monitor inventory and identify potential food waste before it occurs. As a result, business owners often depend on manual checks and staff reminders, which can be inconsistent and time-consuming.

Addressing this challenge is important for improving business profitability, reducing avoidable food waste, and encouraging more sustainable food inventory management practices among SMEs.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ExpiryGuard AI is an automated, AI-assisted food waste prevention application designed to help food-related SMEs, including cafés, restaurants, food vendors, and cloud kitchens, manage inventory efficiently and reduce avoidable food waste.

The application centralises ingredient inventory information, including quantities, expiry dates, costs, categories, and delivery records. Its **automated expiry monitoring** identifies ingredients approaching expiry, enabling business owners to take timely action before stock goes to waste.

A key feature is the **Waste-aware Reorder Scan**, which evaluates existing stock and delivery information to identify potential overstocking and unnecessary reordering. This helps businesses make more informed purchasing decisions, control inventory costs, and minimise food waste.

ExpiryGuard AI also incorporates **automated daily planning and follow-up checks** to support consistent inventory management. These workflows help identify items requiring attention, review outstanding actions, and ensure that important inventory issues are not easily overlooked. **In-app alerts** notify users about upcoming expiry dates, urgent inventory issues, and workflows awaiting approval, allowing them to respond directly within the application.

Our team developed ExpiryGuard AI using **Replit**, a development platform introduced during the SDC Phase 1 workshops. We leveraged AI-assisted development to accelerate prototyping and implementation while applying skills in application design, inventory data management, and automated workflow integration.

By combining automated planning, expiry monitoring, waste-aware restocking support, follow-up checks, and timely in-app alerts, ExpiryGuard AI aims to help SMEs reduce avoidable food waste, improve operational efficiency, and adopt more sustainable inventory management practices.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

ExpiryGuard AI is developed using **Replit**, introduced during the SDC Phase 1 workshops, as our application development environment. We use AI-assisted development to accelerate prototyping, implement application features, and integrate automated workflows.

The prototype consists of the following components:

- **Application development and hosting:** Replit for developing, running, and preparing the application for deployment.
- **User interface:** An interactive dashboard for viewing inventory, expiry risks, reorder recommendations, daily plans, and in-app notifications.
- **Inventory data management:** Structured inventory records containing ingredient IDs, names, quantities, units, expiry dates, costs, categories, and delivery information.
- **Automated workflow logic:** Expiry monitoring, Waste-aware Reorder Scan, automated daily planning, and follow-up checks to identify inventory issues and outstanding actions.
- **Notification system:** In-app alerts to notify users about upcoming expiry dates, urgent inventory issues, and workflows requiring approval.
- **Database and persistence:** The application's existing database stores and retrieves inventory, workflow, and notification records.
- **External APIs:** No additional external API is required for basic expiry monitoring and rule-based workflows. Any GenAI API integration will depend on the AI services actually implemented in the prototype.

System Architecture

ExpiryGuard AI connects inventory data, automated processing, and the user interface in one workflow. Inventory records are processed by the monitoring and decision logic, which identifies potential waste and generates relevant alerts and recommendations. Users can review the results and respond to actions requiring approval through the application.

SDC Phase 1 Workshop Application

The project applies skills gained from the SDC Phase 1 workshops, particularly Replit-based AI-assisted application development, application prototyping, data management, and workflow integration. These skills

enable us to build a functional prototype that addresses inventory inefficiencies and food waste among food-related SMEs.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking principles by focusing on the practical inventory management challenges faced by food-related SMEs, including cafés, restaurants, food vendors, and cloud kitchens.

We considered the difficulties business owners face when tracking ingredient expiry dates, managing stock levels, and preventing unnecessary food waste while handling daily operations.

We identified the key problem as the lack of accessible, timely, and consistent inventory monitoring and follow-up processes, which can lead to avoidable food waste and financial losses.

We developed ExpiryGuard AI around automated expiry monitoring, Waste-aware Reorder Scan, automated daily planning, follow-up checks, and in-app alerts. These features aim to help users identify risks, make informed restocking decisions, and respond to important inventory issues.

Using Replit, introduced during the SDC Phase 1 workshops, we built a functional application prototype with inventory data management, automated workflows, and in-app notifications.

We review the prototype's functionality and workflow outputs to identify issues and improve usability. Before Phase 2 deployment, we aim to validate alert accuracy, verify workflow reliability, test with representative SME inventory scenarios, and refine the interface based on user feedback.

This iterative approach helps us prioritise practical business needs while preparing ExpiryGuard AI for a more reliable and user-friendly deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**UN SDG Target 12.3:** By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses. ExpiryGuard AI supports this target by helping food-related SMEs identify ingredients approaching expiry and make more waste-aware restocking decisions.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Baseline for prototype testing: 0% reduction in expired-stock value** compared with the same simulated inventory scenario without ExpiryGuard AI. This is a simulated comparison baseline, not a measured statistic from real businesses. Real-world food waste levels will require further data collection.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Percentage reduction in the value of inventory that expires unused**, compared with the baseline scenario without ExpiryGuard AI. Calculation: Percentage reduction = [(Baseline expired-stock value − ExpiryGuard expired-stock value) ÷ Baseline expired-stock value] × 100%

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Achieve at least a 20% reduction in the value of expired-unused inventory** across defined simulated inventory test scenarios, compared with the baseline without ExpiryGuard AI, by the final prototype presentation.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

22 October 2026

[Date]